Does Language Really Help a Quantum Portfolio Optimizer?

A Bias-Controlled Study and a Training-Horizon Confound in LLM-Conditioned QAOA

QNLPAI 2026 Student Poster Session · author details withheld for double-blind review

The Central Question

When a backtest says "LLM news-views help a quantum portfolio optimizer," is that **signal — or an artifact?** We build a deliberately bias-controlled pipeline to catch ourselves.

Experimental Pipeline

01

Universe Selection

Dow 30, CRSP total returns — objective, fixed universe

02

LLM Conviction Generation

LLM reads RavenPack headlines time-stamped **strictly before** each decision date → per-asset ±10% view tilts

03

QAOA Cost Hamiltonian

Conviction tilt applied to QAOA cost function → hold 5 of 14 assets, evaluate held-out next quarter

04

Controls Applied

Matched random-tilt placebo; ablation vs. RavenPack sentiment; **two LLMs on identical headlines**

The Two-LLM Design

qwen2.5-14B

Training overlaps 2024 — **contaminated**

Llama 2

Sept-2022 cutoff — genuinely **out-of-sample**

Both models read the *same* RavenPack headlines. Only the training-horizon relationship to the backtest period differs.

⚠ **Key Finding — The apparent "language signal" is the LLM's hindsight:** The out-of-sample alpha collapses to indistinguishable-from-random with a genuinely out-of-sample model.

+0.76% per quarter

qwen2.5-14B Premium

Over random tilts (p≈0) — training overlaps 2024 backtest window

−0.23% per quarter

Llama 2 Premium

Over random tilts (p=0.94) — Sept-2022 cutoff, indistinguishable from random

47–68%

Optima Changed

A ±10% view changes the exact optimum in 47–68% of instances — language has real mechanism leverage on the selection.

7.7%

True Optimum Found

Shallow QAOA is a weak exact optimizer — no quantum-advantage claim is made

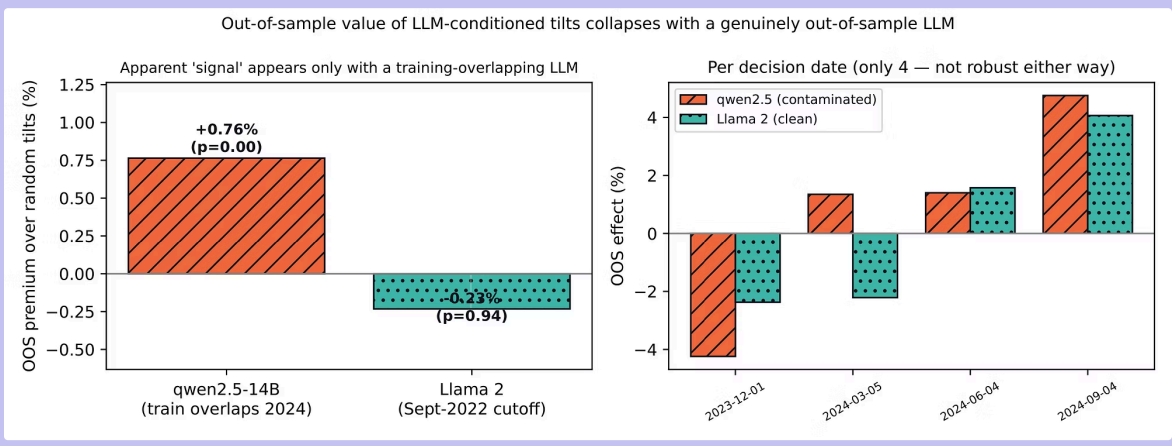


FIGURE 1 — Out-of-sample value of LLM-conditioned tilts collapses with a genuinely out-of-sample LLM.

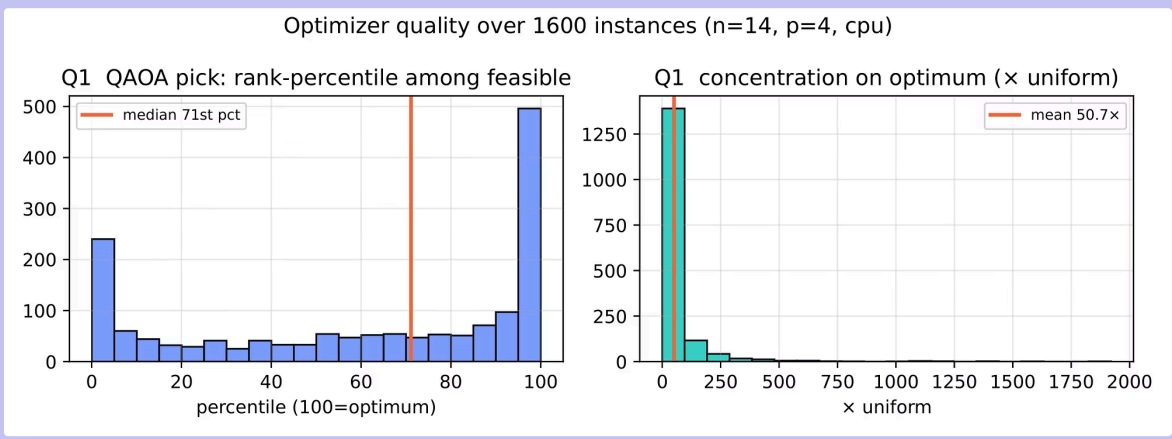


FIGURE 2 — QAOA optimizer quality over 1600 instances (n=14, p=4): rank-percentile among feasible portfolios (median 71st) and concentration on the optimum (mean 50.7× uniform).

Honest Limitations

- Only **4 decision dates** — instance-level p is pseudo-replicated
- Clean and contaminated LLMs differ in **capability** as well as cutoff date
- Small-n **exact simulation**; large-cap US equities only; no transaction costs
- None of these limitations **inflate** the central result

Contribution

→ A **reproducible bias-control template** for LLM-conditioned financial backtests

→ A **quantified warning**: date your LLM against your backtest window, or training-horizon look-ahead will manufacture alpha that isn't there

✓ **Reproducibility:** Fixed seeds, committed CRSP + conviction caches; code released upon publication.